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Photodynamic effects of two hydroxyanthraquinones

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Abstract

The aim of this work was to investigate the photodynamic action of electron-rich anthraquinones, viz., cynodontin (CYN) and cynodontin-5,8-dimethylether (CYNM). Both optical and EPR methods are used to detect the generation of singlet oxygen. Based on RNO bleaching, relative to rose bengal (RB), singlet oxygen generating efficiencies of CYN and CYNM are derived to be 0.055 and 0.254, respectively. The formation of superoxide anion via electron transfer to O_2 was monitored by optical spectroscopy, using SOD-inhibitable cytochrome *c* reduction assay. The production of $O_2^{-\cdot}$ is enhanced in the presence of electron donors such as EDTA and NADH. Photolysis of CYN and CYNM in DMSO, in the presence of 5,5-dimethyl-1-pyrroline-*N*-oxide (DMPO), generates a multi-line EPR spectrum, characteristic of spin adduct mixture of $O_2^{-\cdot}$ and $^{\cdot}OH$. Both optical and ESR measurements indicate that $O_2^{-\cdot}$ (Type I) and $^1O_2$ (Type II) paths are involved in CYN and CYNM photodynamic action.
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Keywords: Anthraquinone; 5,5-dimethyl-1-pyrroline-*N*-oxide (DMPO); Singlet oxygen; Superoxide anion; Spin trapping

1. Introduction

A large number of quinones have been tested for antitumor activity and now quinones form one of the largest classes of antitumor agents approved for clinical use [1,2]. Clinically used mitomycin and streptonigrin possess *p*-benzoquinone moiety with heterocyclic groups, whereas the anthracyclines, doxorubicin and daunorubicin consist of anthraquinone moiety [3]. Photoinduced production of superoxide anion radicals by adriamycin and daunorubicin is well known [4]. Phleichrome, a quinoid agent has been shown to photogenerate $^1O_2$ and $O_2^{-\cdot}$ [5]. There is considerable interest to understand the factors contributing to the photodynamic activity of the anthraquinones [6–9]. Many antitumor aminoanthraquinone derivatives have been studied for their efficiency to generate reactive oxygen species upon visible light illumination [10–12]. The photochemical reactivity of anthraquinone derivatives is reported to increase with singlet–triplet intersystem crossing efficiency [13].

We have undertaken a systematic investigation of photodynamic action of various natural as well as synthetic quinones [14,15]. The naturally occurring anthraquinone cynodontin (CYN) is available from heartwood of *Maesopsis emini* (Rhamnaceae). CYN can also be synthesised by condensing 3,6-dimethoxyphthalic anhydride and toluhydroquinone in anhydrous aluminium chloride, sodium chloride melt [16]. Hydroxyl substituted anthraquinones are found to show mutagenicity [17]. The methylated product of CYN is the cynodontin-5,8-dimethylether (CYNM). In this paper, we investigate the role of –OH (hydroxy) and –OMe (methoxy) substituents in the anthraquinone moiety on the photodynamic effect, under aerobic conditions.

2. Material and methods

2.1. Chemicals

The chemical structures of the quinones studied are given in Fig. 1. These quinones, CYN and CYNM were received as gift from the Department of Natural Products Chemistry, Madurai Kamaraj University, Madurai, India. UV–VIS

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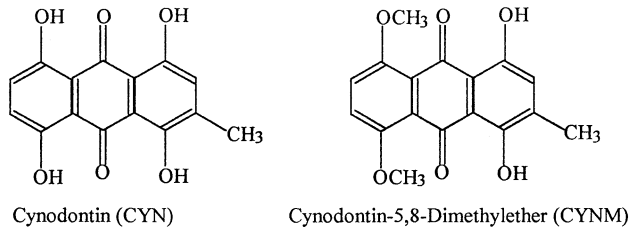

Fig. 1. Chemical structures of CYN and CYNM.

spectra of CYN and CYNM recorded in Shimadzu UV–VIS spectrometer (UV-160), are shown in Fig. 2.

Superoxide dismutase (SOD), catalase and cytochrome *c* were purchased from Sigma Chemical Co., while reduced nicotinamide adenine dinucleotide (NADH) was obtained from Boehringer Mannheim. *N*,*N*-Dimethyl-4-nitroso aniline (RNO), 1,4-diazabicyclo[2,2,2]octane (DABCO), diethyltriaminopentaacetic acid (DETAPAC), and rose bengal (RB) were obtained from Aldrich. The spin trap 5,5-dimethyl-1-pyrroline-*N*-oxide (DMPO) was obtained from Aldrich and was purified using activated charcoal [18]. 2,2,6,6-Tetramethyl piperidinol (TEMPL) and 4-hydroxy-2,2,6,6-tetramethylpiperidine-1-oxyl (TEMPOL) was obtained from Merck. Imidazole, ethylenediaminetetraacetic acid (EDTA) and sodium azide were purchased from S.D. Fine Chemicals, India. Imidazole was used after repeated crystalisation from doubly distilled water. All the other reagents used here were of analytical grade.

2.2. Light source

The reaction mixture in a 1 cm spectrometric cuvette, placed at a distance of 12 cm, was continuously irradiated using 150 W xenon lamp. A filter combination of 10 cm potassium iodide solution (1 g in 100 ml) and 1 cm pyridine was used as a bandpass filter to achieve a spectral window of 300–700 nm. The irradiation was carried out in an open cuvette in equilibrium with air.

2.3. EPR spin trapping

The EPR spin trapping experiments were carried out using JEOL TES-TE 100 ESR spectrometer, operating at X-band frequency with 100 kHz field modulation at room temperature as per the details published elsewhere [15]. Solution of quinone (200 μM) and DMPO (100 mM) in DMSO, placed in a quartz cuvette, was continuously stirred using a magnetic stirrer during irradiation. Irradiated solution was drawn into a gas-permeable Teflon capillary tube (0.8 mm inside diameter, 0.5 mm wall thickness). The Teflon tube was folded, inserted into a narrow quartz tube, and placed in the EPR cavity. The microwave power was 2 mW and modulation amplitude was kept at 0.5 G. The spectral identification was confirmed by independently simulating the spectra with known hyperfine coupling constants and comparing them with the experimentally obtained EPR spectrum. To simulate the EPR spectra, a BASIC computer program was used. The output from this program was plotted to get the simulated spectra.

The photogeneration of $^1O_2$ by quinone in DMSO was also investigated by EPR measurement. TEMPL was used as a singlet oxygen trapper by converting it to an EPR-detectable nitroxide free radical [19,20]. Air saturated reaction mixture (1 ml) containing 0.02 M TEMPL and 0.2 mM of quinones was irradiated and the increase in EPR signal intensity was followed as a function of irradiation time.

2.4. Optical spectroscopy

Optical assays for $^1O_2$ and $O_2^{-\cdot}$ were performed on a Shimadzu UV–VIS spectrometer (UV-160) as well as on Specord S100 UV–VIS spectrometer Analytik jena AG, Jena, Germany. The RNO-bleaching method was used to determine the $^1O_2$-generating efficiency [21]. The quinone solution (200 μM) was irradiated in the presence of imidazole (10 mM) and RNO (50 mM) in a 50 mM phosphate buffer (pH = 7.4). The optical assay is studied with a total volume of 2 ml buffer medium. The concentrations of compounds, CYN and CYNM are 1 mM and 200 μM,

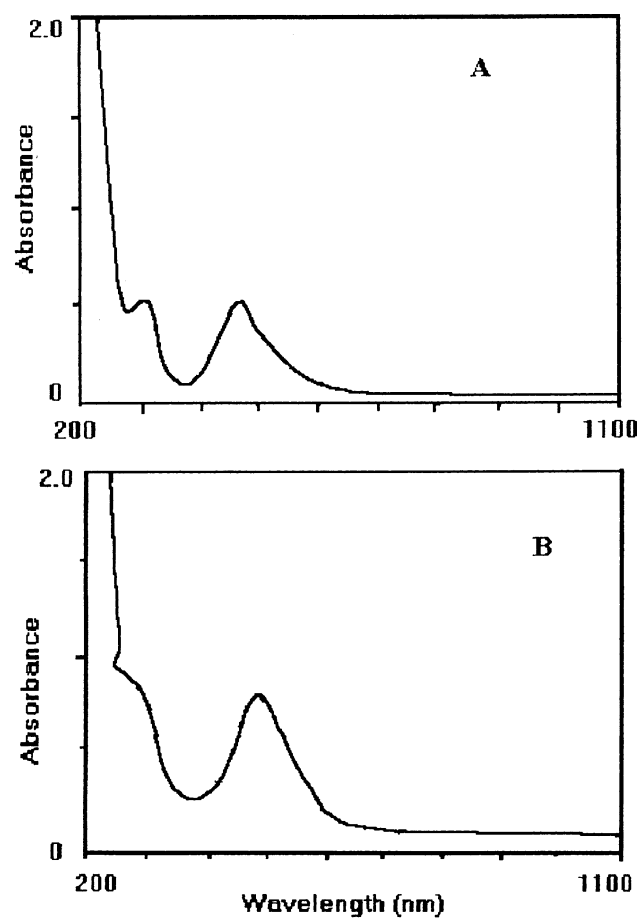

Fig. 2. UV–VIS spectra of (A) CYN (1 mM) λ_{max} at 306 and 462 nm, and (B) CYNM (200 μM) λ_{max} at 484 nm recorded in phosphate buffer (pH = 7.4).

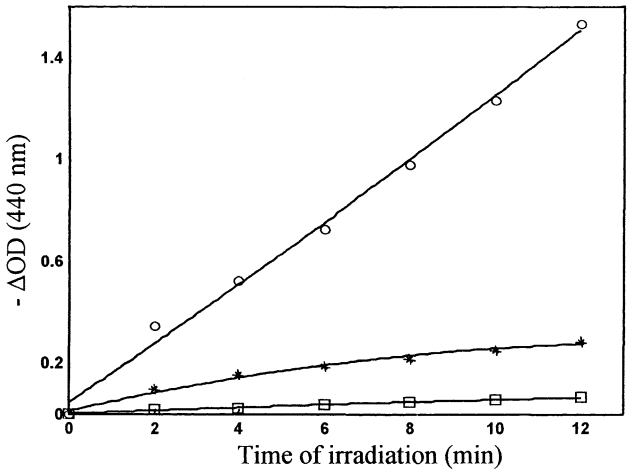

Fig. 3. Photosensitised RNO bleaching measured at 440 nm in the presence of imidazole (10 mM) in 50 mM phosphate buffer (pH = 7.4) with RB (0.015 mM) (○○○), CYNM (***) and CYN (□□□), as a function of irradiation time.

respectively. A plot of concentration in the range of (0.2–1 mM) versus absorbance, shows a linear relationship. The bleaching of RNO by $^1O_2$ was followed spectrophotometrically by observing the decrease in the 440 nm absorption peak of RNO as a function of irradiation time. The rate of disappearance of quencher obeys first-order kinetics [22] and the slope of the first-order plot was calculated as described earlier [14]. The quinones under investigation along with the reference singlet generator, RB (0.015 mM) were studied under identical conditions. The relative ratio of slopes corrected for the light absorption and photon energy [23] gives the $^1O_2$ quantum yield. The reference RB $^1O_2$ quantum yield was taken to be 0.76 [24]. Interference of $O_2^{-\cdot}$ and H_2O_2 in the RNO bleaching was eliminated by adding 50 μg/ml of SOD and catalase.

Generation of superoxide was also detected by using SOD-inhibitable cytochrome *c* reduction method [25]. Quinones were illuminated in the presence of ferricytochrome *c* (40 μM) in 50 mM phosphate buffer (pH = 7.4). The reduction was followed by observing the increase in the 550 nm absorption peak of cytochrome *c* as a function of irradiation time. The $\Delta\varepsilon_{550} = 20000\ M^{-1}\ cm^{-1}$ was used for the reduced-oxidised cytochrome *c* [25].

2.5. Cyclic voltammetry

Redox potentials of acetonitrile solutions of quinones (1 mg; 5 ml acetonitrile) were measured by cyclic voltammetry technique in BAS 50A electrochemical analyzer (Bioanalytical Systems, West Lafayette, IN) using glassy carbon working electrode, platinum auxiliary electrode and Ag/AgCl reference electrode. Glassy carbon electrode was resurfaced with alumina before use. Tetra-*n*-butylammonium perchlorate (TBAP) (50 mM) was used as the supporting electrolyte. Each solution was purged with nitrogen for 10 min prior to measurement and the cyclic voltammograms were recorded under nitrogen atmosphere.

3. Results and discussion

3.1. Type II mechanism

Quinones are known to participate both in Type I and Type II photochemical reactions [26,27]. For both CYN and CYNM, bleaching of RNO was not observed in dark or in the absence of oxygen. Fig. 3 shows the loss of RNO absorbance at 440 nm versus irradiation time. The singlet oxygen-generating yields, evaluated as described earlier, are 0.055 and 0.254 for CYN and CYNM, respectively. RNO has been successfully applied in a wide variety of photochemical systems as a scavenger of $^{\cdot}OH$ radical. In our experimental conditions, RNO is photochemically inert, because the radicals produced, $^{\cdot}OH$ and $O_2^{-\cdot}$ are scavenged by the added catalase and SOD. Additional support for the generation of $^1O_2$ was obtained by studying the effect of specific $^1O_2$ inhibitors such as DABCO and sodium azide on RNO bleaching. The results are similar for both compounds, and hence the results are shown in Fig. 4 only for CYNM. The singlet oxygen quenching rate constants of imidazole ($2 \times 10^7\ M^{-1}\ s^{-1}$) and DABCO ($1.5 \times 10^7\ M^{-1}\ s^{-1}$) are comparable [28]. Since the quenching rate constants of equimolar amounts (10 mM) of imidazole and DABCO are similar, both of these compounds would decrease the observed rate by about half, when compared to the rate of RNO bleaching in the absence of DABCO [29]. In this experiment, the measurements are carried out in the presence of imidazole (10 mM) and DABCO (5 mM), so the reduction rate is decreased by about 25% (0.0304), when compared to the rate of RNO bleaching in the absence of

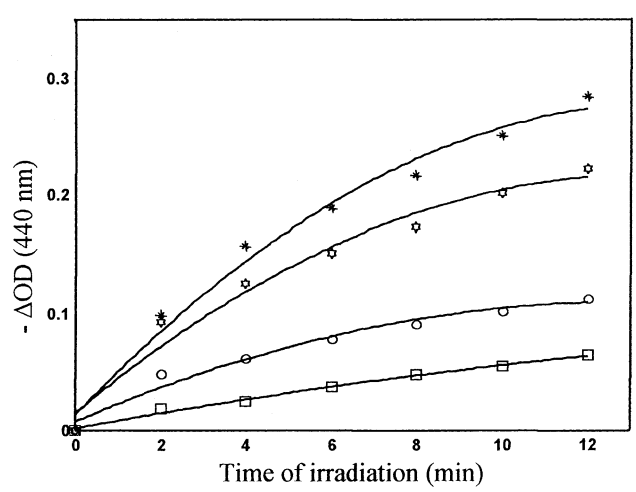

Fig. 4. Photosensitised RNO bleaching measured at 440 nm in the presence of imidazole (10 mM) in 50 mM phosphate buffer (pH = 7.4) with CYNM (***) and CYN (□□□), as a function of illumination time. Inhibition of photosensitised RNO bleaching of CYNM by 0.1 mM sodium azide (○○○) and 5 mM DABCO (✡✡✡) is also presented.

DABCO. Similarly, sodium azide reacts with $^1O_2$ about 100 times faster than imidazole. But in the present study, the concentrations of sodium azide and imidazole used are 0.1 and 10 mM, respectively. Hence, at these concentrations, it was found that the RNO bleaching was inhibited by 50% (0.0151). These results confirm the generation of $^1O_2$ during the photodynamic process.

The optical RNO bleaching method has a limitation, if compounds and RNO have the same wavelength maxima. Hence, another technique, the EPR-TEMPL method was used to study the $^1O_2$ yield. In EPR technique, the problems related to optical study are eliminated. The conversion of TEMPL to TEMPOL is due to $^1O_2$. When the reaction mixture of TEMPL and quinones was irradiated, an EPR spectrum of three equal intensity lines, characteristic of a nitroxide radical (TEMPOL), was detected. The analytical grade sample TEMPOL was purchased from chemical company Merck. The EPR spectrum was recorded for this commercially procured TEMPOL in DMSO. The hyperfine splitting constant ($A_N = 15.6$ G) of the nitroxide radical formed was found to be identical with that of the commercial sample. EPR signal intensity of TEMPOL increased with an increase of irradiation time as shown in Fig. 5. Under similar conditions, in the absence of quinones, negligible amounts of nitroxide free radical were formed. Control experiments indicated that quinones, oxygen and light were all essential for the production of TEMPOL, indicating that the formation of the nitroxide radical is a photodynamic process. The EPR signal was suppressed by $^1O_2$ quencher sodium azide (Fig. 5) during CYNM photosensitisation, confirming the formation of $^1O_2$.

The generation of TEMPOL by quinones was found to be parallel to their $^1O_2$ quantum yield. From Fig. 5, the $^1O_2$ generating efficiency ratio of RB, CYN and CYNM were determined to be 1:0.07:0.33, respectively, consistent with the values determined by the RNO bleaching method. The

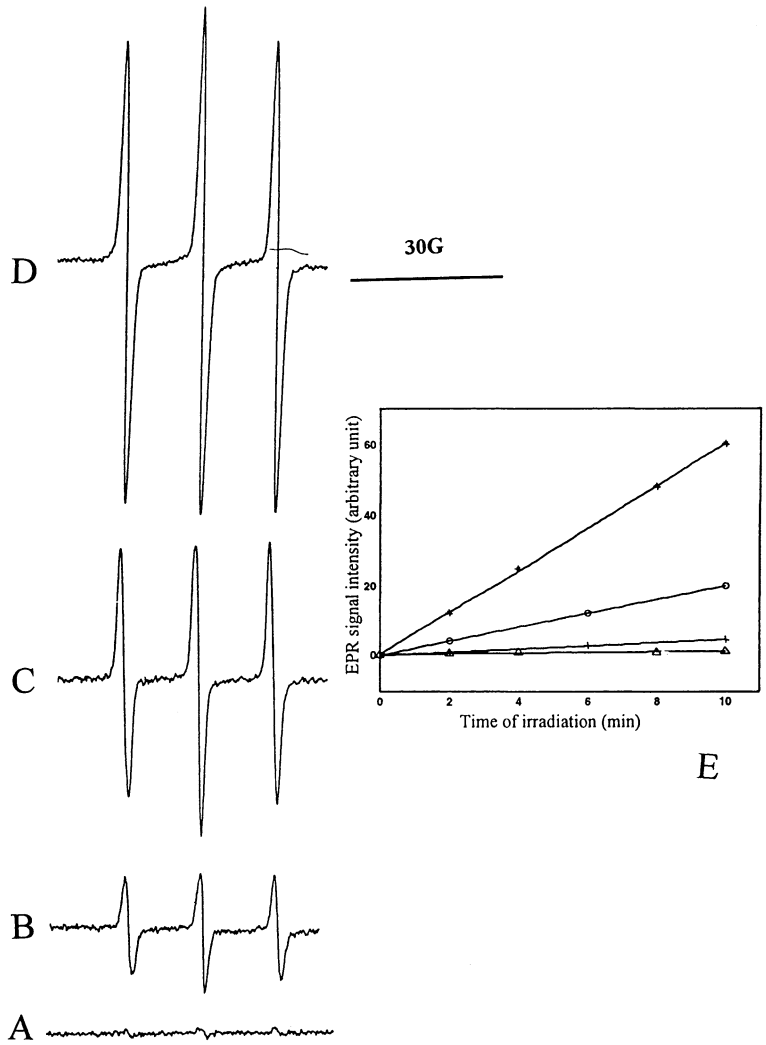

Fig. 5. The EPR spectra of TEMPOL generated during the photoirradiation of DMSO solution of CYNM (200 μM) in the presence of TEMPL (20 mM) at 300 K. (A) In the dark, (B) 2-min irradiation, (C) 6-min irradiation, (D) 10-min irradiation. Panel E presents the formation of TEMPOL in the presence of RB (***), CYN (+++), CYNM (○○○) and the inhibitory effect of 1mM sodium azide (△△△) on the formation of TEMPOL radical during photoirradiation of CYNM. EPR spectrometer settings: microwave power 2 mW, modulation amplitude 1 G, time constant 0.1 s, scan rate 4 min.

EPR-TEMPL and RNO bleaching experiments were carried out in DMSO and buffer solvents, respectively. The solvent DMSO used is analytical grade. It is likely to contain 2% water. Hence, it may also form a hydrogen bond. Hence, the singlet oxygen-generating efficiencies are almost similar in both methods [26,30].

3.2. Type I mechanism

The photoexcited states of quinones are known to be quenched by molecular oxygen to generate reactive oxygen species (Type I). Hence, the formation of $O_2^{-\cdot}$ during the photoillumination of quinones, was studied by following the SOD-inhibitable reduction of ferricytochrome *c*. The rate of cytochrome *c* reduction, when air-saturated solutions of quinones were photolysed in the presence of EDTA and cytochrome *c* (40 μM) in phosphate buffer (pH = 7.4) was studied. The rates of superoxide anion formation by CYN and CYNM were determined to be 0.003 and 0.010 μM/s, respectively. The addition of SOD (50 μg/ml) inhibited the cytochrome *c* reduction. Although cytochrome *c* may be reduced by a variety of electron carriers, measuring only the reduction that is inhibitable by SOD makes the measurements more specific for superoxide anion. Additionally, the effect of electron donor NADH, on $O_2^{-\cdot}$ production was also studied. Fig. 6 shows the effect of NADH on the SOD-inhibitable cytochrome *c* reduction by CYN and CYNM. The addition of an electron donor like NADH enhances the reduction rate and it is readily seen from the graph. Hence, the numerical data are not included. The enhancement of the rate of reduction by electron donors, suggests that photoreduced quinone anion may be the precursor for the formation of $O_2^{-\cdot}$.

Photogeneration of superoxide anion in CYNM was further verified by EPR spin trapping technique. EPR studies were carried out in DMSO in which DMPO-$O_2^{-\cdot}$ adduct has longer lifetime [31]. EPR signal was not observed when DMPO alone was irradiated. Also, no EPR signal was observed in the dark (Fig. 7A). A multi-line EPR spectrum (Fig. 7B) consisting of DMPO/OH and DMPO/OOH spin adducts appeared upon photoirradiation of the sample consisting of CYNM (200 μM) and DMPO in air-saturated DMSO. The observed EPR spectrum could be readily analysed in terms of a mixture of two type of spin adducts. By using computer simulation, the EPR spectra of these two spins adducts were simulated separately with their corresponding hyperfine coupling constants. When these two spectra were combined in the ratio of 6:4 for $O_2^{-\cdot}$ and $^{\cdot}OH$, respectively, the simulated EPR spectrum (Fig. 7C) matched well with the experimentally observed one. Hyperfine coupling constant values for one of the spin adducts was obtained, from simulation, as a nitrogen triplet (A_N = 13.0 G) split by a proton (A_H^{β} = 10.3 G), which in turn is further split by a secondary proton (A_H^{γ} = 1.3 G). The hyperfine coupling constants of the other spin adduct are A_N = 14.2 G and A_H^{β} = 11.6 G. These splitting parameters of DMPO adducts could be readily assigned to DMPO-$O_2^{-\cdot}$ and DMPO-OH, respectively, based on the literature data [32]. The addition of SOD (50 μg/ml) prior to illumination prevented the formation of both the DMPO-$O_2^{-\cdot}$ and

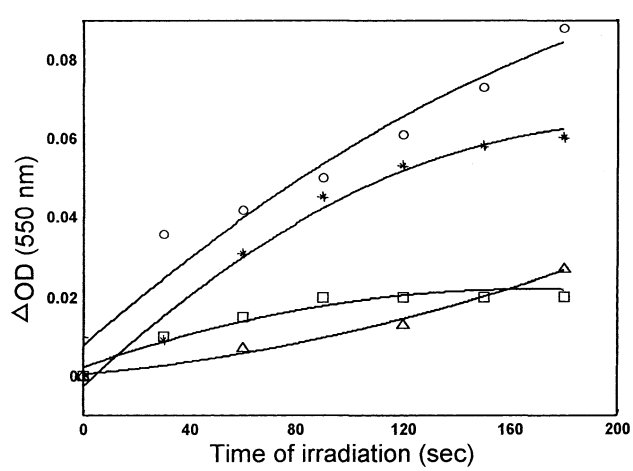

Fig. 6. Photosensitised cytochrome *c* reduction in the presence of NADH by sensitisers, CYN (○○○), CYNM (***), CYNM + SOD (50 μg/ml) (□□□) and in the absence of NADH by CYN (△△△).

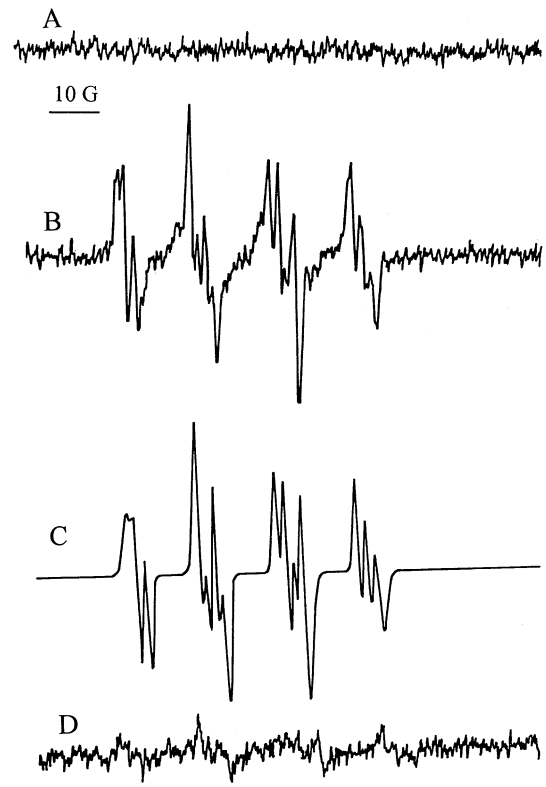

Fig. 7. EPR spectra of DMPO adduct in aerated DMSO solution containing CYNM (200 μM) and DMPO (100 mM). (A) In the dark, (B) after 4-min irradiation. The composite computer-simulated EPR spectrum (C) contains two spin adducts: DMPO-$O_2^{-\cdot}$ (A_N = 13.0 G, A_H^{β} = 10.3 G, A_H^{γ} = 1.3 G) and DMPO-OH adduct (A_N = 14.2 G and A_H^{β} = 11.6 G) in the ratio of 6:4. (D) The same as B, in the presence of SOD (50 μg/ml). Spectrometer settings: microwave power, 2.0 mW; modulation frequency, 100 kHz; modulation amplitude, 0.5 G; time constant, 0.1 s; scan rate, 4 min; scan width, 200 G.

DMPO-OH signals (Fig. 7D). These observations suggest the formation of DMPO-$O_2^{-\cdot}$ as the primary product, which may later decompose to form the observed DMPO-OH, adduct. The electron-donating substituents in quinone (CYNM) increase the yield. Since the spectral behaviour of CYN and CYNM are similar, the results of only one compound, viz., CYNM are presented in Fig. 7.

3.3. Redox potentials

Electrochemical studies were carried out to determine the reduction potential for both CYN and CYNM. Cyclic voltammograms of the two quinones were measured in the range of +0.5 to −2.0 V. The reduction potentials for CYN and CYNM were found to be −0.805 and −0.813 V, respectively, indicating that both anthraquinones, potentially, could be reduced. As shown in Fig. 8, the reduction of CYN and CYNM is quasireversible.

The influence of the substitution on the quinone ring by hydroxyl and methoxyl groups is demonstrated by comparison of anthraquinones CYN and CYNM. Electron-donating substituents cause a shift of $E_{1/2}$ to more negative value (Table 1), whereas electron-withdrawing substituents cause opposite effect [33]. CYNM contains a methoxy group, causing a shift to more negative values than CYN because of their electron-donating properties [34].

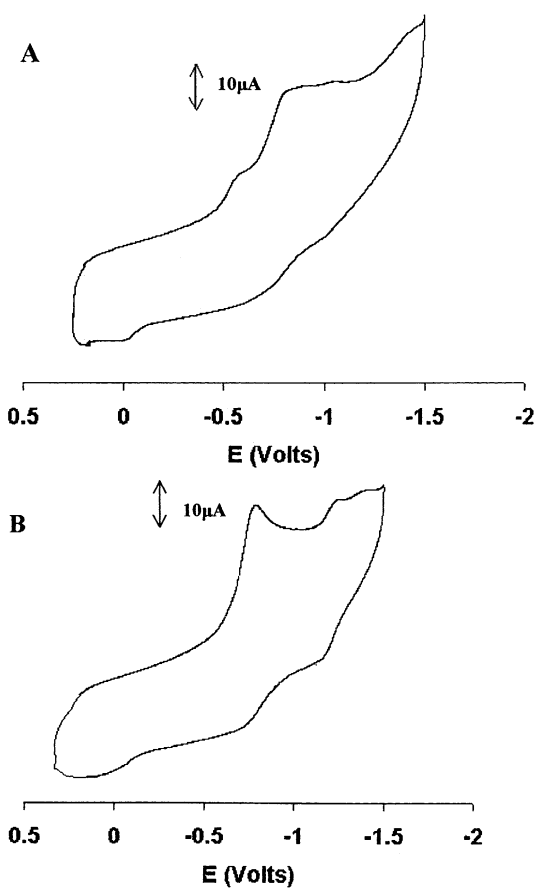

Fig. 8. Cyclic voltammograms of (A) CYN and (B) CYNM in acetronitrile containing tetra-*n*-butylammonium perchlorate as supporting electrolyte (0.05 M) at 100 mV/s scan rate.

Table 1
Cyclic voltammetric data[a] and $^1O_2$-generating efficiencies[b] of quinones

Substrate	E_{pc}	E_{pa}	ΔE_p	$E_{1/2}$	$\Phi\ ^1O_2$	$O_2^{-\cdot}$ (µM/s)	Reference
CYN	−0.805	−0.662	0.143	−0.734	0.055	0.003	this work
CYNM	−0.813	−0.712	0.101	−0.763	0.254	0.010	this work
BQ-I	−1.250	−1.125	0.125	−1.183	0.180	0.001	[15]
BQ-II	−1.245	−1.111	0.134	−1.178	0.070	0.001	[15]
MD	−0.760	−0.700	0.060	−0.730	0.130	0.036	[14]
TP	−0.620	−0.570	0.050	−0.595	0.300	0.038	[14]

E_{pc} and E_{pa}—electrode potential of cathode and anode.
ΔE_p—change in potential.
$E_{1/2}$—half-wave potential.
$\Phi\ ^1O_2$—singlet oxygen quantum yield.
$O_2^{-\cdot}$ (µM/s)—rate of formation of superoxide.
[a] Potentials in volts against Ag/AgCl; scan rate, 100 mV/s.
[b] With reference to singlet oxygen quantum yield of RB taken as 0.76.

3.4. Structure–activity correlation

Our results show that the photodynamic efficiency of CYNM, to generate both $^1O_2$ and $O_2^{-\cdot}$, is higher than that of CYN. This may be due to the electron-donating (methoxy) groups directly attached to the quinone ring system. This is in accordance with the report that electron-donating functional groups such as amino and alkoxy groups enhance the

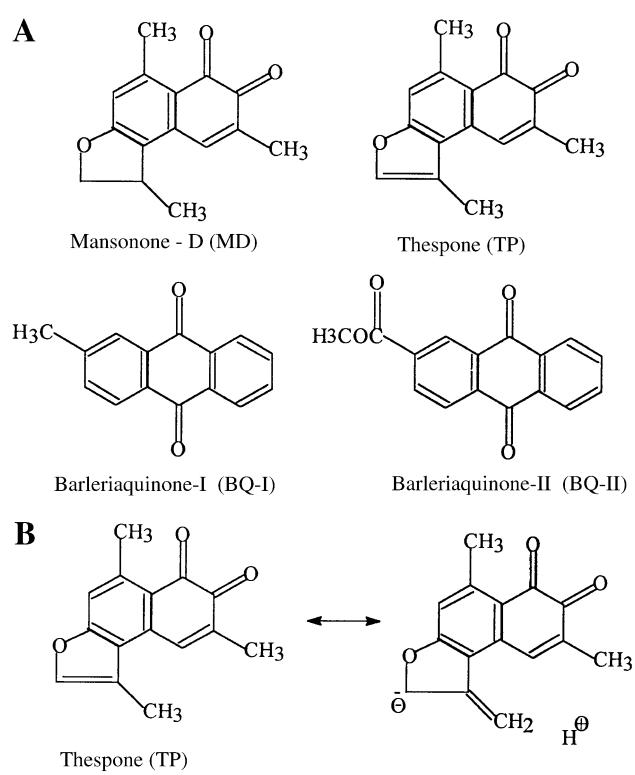

Fig. 9. (A) Chemical structure of quinones MD, TP, BQ-I and BQ-II. (B) Hyperconjugative effect of methyl group attached to heterocyclic ring system of TP.

photodynamic efficiency [35]. Our earlier studies have also revealed a strong relationship between substituent effect and singlet oxygen quantum yields. For example, the two anthraquinones barleriaquinone-I and -II (BQ-I and BQ-II) have structural difference in the seventh position (Fig. 9). BQ-I has an electron releasing methyl group, but BQ-II has electron-withdrawing methyl ester at the seventh position [15]. Because of the electron-donating functional group directly attached to the ring, BQ-I shows higher quantum yield of $^1O_2$ than BQ-II (Table 1). However, both BQ-I and BQ-II show almost similar yields of $O_2^{-\cdot}$ in the absence of any electron donor. But in the presence of an electron donor reagent like EDTA, the rate of cytochrome *c* reduction was found to be 0.0035 and 0.0017 μM/s for BQ-I and BQ-II, respectively, showing the influence of the electron-donating group. This may be due to the possibility of involvement of Type I reaction in preference to Type II. Similar effects were also noticed for the quinones, Mansonone-D (MD), and Thespone (TP), which have related structures [14]. The fused heterocyclic ring in both MD and TP has one methyl substitution. But in TP, the methyl group attached to the heterocyclic ring system can donate the electron to the quinone ring due to extensive conjugation as shown in Fig. 9.

3.5. Conclusion

It is concluded that both electron transfer (Type I) and energy transfer (Type II) pathways are involved in the photodynamic process by CYN and CYNM. The compound, CYNM shows higher photodynamic efficiency than CYN, because of the presence of electron-donating −OMe group.

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References

[1] G. Powis, Free radical formation by antitumor quinones, Free Radic. Biol. Med. 6 (1989) 63–101.
[2] H.W. Moore, R. Czerniak, A. Hamdan, Natural quinones as quinone methide precursors—ideas in rational drug design, Drug Expl. Clin. Res. 12 (1986) 475–494.
[3] E. Dichancaite, N. Cenas, A. Kalvelyte, N. Serapiniene, Toxicity of daunorubicin and naphthaquinones to HL-60 cells: an involvement of oxidative stress, Biochem. Mol. Biol. Int. 41 (1997) 987–994.
[4] K.J. Reszka, P. Kolodziejczyk, J.W. Lown, Photosensitization by antitumor agents: 3. Spectroscopic evidence for superoxide and hydroxyl radical production by anthrapyrazole-sensitized oxidation of NADH, Free Radic. Biol. Med. 2 (1986) 267–274.
[5] Z.J. Diwu, J.W. Lown, Photosensitization with anticancer agents. 20-EPR studies on the photodynamic action of phleichrome: formation of semiquinone radical and activated oxygen species on illumination with visible light, Free Radic. Biol. Med. 18 (1995) 357–363.
[6] S. Ray, V. Ponnathpur, Y. Huang, C. Tang, M.E. Mahoney, A.M. Ibrado, G. Bullock, K. Bhalla, 1-beta-D-arabinofuranosylcytosine-, mitoxantrone-, and paclitaxel-induced apoptosis in HL-60 cells: improved method for detection of internucleosomal DNA fragmentation, Cancer Chemother. Pharmacol. 34 (1994) 365–371.
[7] K.J. Reszka, P. Bilski, C.F. Chignell, J.A. Hartley, N. Khan, R.L. Souhami, A.J. Mendonca, J.W. Lown, Photosensitization by anticancer agents: 11. Mechanisms of photosensitization of human leukemic cells by diaminoanthraquinones: singlet oxygen and radical reactions, J. Photochem. Photobiol., B Biol. 15 (1992) 317–335.
[8] A. Andreoni, A. Colasanti, V. Malatesta, P. Riccio, G. Roberti, Enhancement of antitumor drug cytotoxicity via laser photoactivation, Photochem. Photobiol. 53 (1991) 797–805.
[9] T. Shiraki, Y. Sugiura, Visible light induced DNA cleavage by the hybrid antitumor antibiotic dynemicin A, Biochemistry 29 (1990) 9795–9798.
[10] K. Reszka, J.W. Lown, Photosensitization of anticancer agents-8. One-electron reduction of mitoxantrone: an EPR and spectrophotometric study, Photochem. Photobiol. 50 (1989) 297–304.
[11] J.A. Hartley, K. Reszka, J.W. Lown, Photosensitization by antitumor agents: 7. Correlation between anthracenedione-photosensitized DNA damage, NADH oxidation and oxygen consumption following visible light illumination, Photochem. Photobiol. 48 (1988) 19–25.
[12] K.J. Reszka, P. Bilski, C.F. Chignell, J.A. Hartley, N. Khan, R.L. Souhami, A.J. Mendoca, J.W. Lown, Photosensitisation by antitumor agents: 11. Mechanisms of photosensitisation of human leukemic cells by diaminoanthraquinones: singlet oxygen and radical reactions, J. Photochem. Photobiol. 15 (1992) 317–335.
[13] K. Reszka, P. Kolodziejczyk, P.G. Tsoungas, J.W. Lown, Photosensitisation by antitumor agents: 6. Production superoxide radical and hydrogen peroxide during illumination of diaminoanthracendiones in the presence of NADH in aqueous solutions: an EPR study, Photochem. Photobiol. 47 (1988) 625–633.
[14] J.J. Inbaraj, R. Gandhidasan, R. Murugesan, Photodynamic action of some naturally occurring quinones: formation of reactive oxygen species, J. Photochem. Photobiol., A Chem. 124 (1999) 95–99.
[15] J.J. Inbaraj, M.C. Krishna, R. Gandhidasan, R. Murugesan, Cytotoxicity, redox cycling and photodynamic action of two naturally occurring quinones, Biochim. Biophys. Acta 1472 (1999) 462–470.
[16] L. Tikkanen, T. Matsushima, S. Natori, Mutagenicity of anthraquinones in the *Salmonella* preincubation test, Mutat. Res. 116 (1983) 297–304.
[17] V.M. Chari, S. Neelakantan, T.R. Seshadri, Constitution of tectoleafquinone—a synthetic study, Indian J. Chem. 7 (1969) 40–43.
[18] B. Kalyanaraman, C.C. Felix, R.C. Sealy, Photoionisation of melanin precursors: an electron spin resonance investigation using the spin trap 5,5-dimethyl-1-pyrroline-1-oxide (DMPO), Photochem. Photobiol. 36 (1982) 5–12.
[19] Y. Lion, M. Demelle, A. Van De Vorst, New method of detecting singlet oxygen production, Nature 263 (1976) 442–443.
[20] J. Moan, E. Wold, Detection of singlet oxygen production by ESR, Nature 279 (1979) 450–451.
[21] I. Kraljic, S. El Mohsni, A new method for the detection of singlet oxygen in aqueous solution, Photochem. Photobiol. 28 (1978) 577–581.
[22] E. Gandin, Y. Lion, A. Van de Vorst, Quantum yield of singlet oxygen production by Xanthene derivatives, Photochem. Photobiol. 37 (1983) 271–278.
[23] E. Gandin, Y. Lion, J. Photochem. 30 (1982) 77–81.

[24] P.C.C. Lee, M.A.J. Rodgers, Laser flash photokinetic studies of rose bengal sensitised photodynamic interactions of nucleotides and DNA, J. Photochem. Photobiol. 45 (1987) 79–86.
[25] W.H. Koppenol, J. Butler, The radiation chemistry of cytochrome *c*, Isr. J. Chem. 24 (1984) 11–16.
[26] T. Wu, J. Shen, A. Song, S. Chen, M. Zhang, T. Shen, Photodynamic action of amino substituted hypocrellins: EPR studies on the photogeneration of active oxygen and free radical species, J. Photochem. Photobiol., B Biol. 57 (2000) 14–21.
[27] Z. Diwu, J.W. Lown, Hypocrellins and their uses in photosensitisation, Photochem. Photobiol. 52 (1990) 609–616.
[28] E. Wilkinson, J.G. Brummer, Rate constants for the decay and reactions of the lowest electronically excited singlet state of molecular oxygen solution, J. Phys. Chem. Ref. Data 10 (1981) 809–999.
[29] I. Rosenthal, C.M. Krishna, P. Riesz, E. Ben-Hur, The role of molecular oxygen in the photodynamic effect of phthalocyanines, Radiat. Res. 107 (1986) 136–142.
[30] Z. Diwu, J.W. Lown, Photosensitization with anticancer agents: 15. Perylenequinoid pigments as potential photodynamic therapeutic agent: formation of semiquinone radicals and reactive oxygen species on illumination, J. Photochem. Photobiol., B Biol. 18 (1993) 131–143.
[31] E. Ben-Hur, A. Carmichael, P. Riesz, I. Rosenthal, Photochemical generation of superoxide anion radical and the cytotoxicity of phthalocyanines, Int. J. Radiat. Biol. 48 (1985) 837–846.
[32] G.R. Buettner, Spin trapping: ESR parameters of spin adducts, Free Radic. Biol. Med. 3 (1987) 259–303.
[33] R.J. Driebergenm, W. Verboom, D.N. Reinhoudt, P. Lelieved, Electrochemistry of potential bioreductive alkylating quinones: its use in the development of new aziridinylquinones, Anticancer Res. 6 (1986) 605–620.
[34] C. Hansch, A. Leo, S.H. Vnger, K.H. Kim, D. Nikaitaini, E.J. Lien, Aromatic substituent constants for structure–activity relationship, J. Med. Chem. 16 (1973) 1207–1216.
[35] J.S. Driscoll, G.F. Hazard Jr., H.B. Wood Jr., A. Goldin, Structure-antitumor activity relationships among quinone derivatives, Cancer Chemother. Rep., Part 2 4 (2) (1974) (April).